batmonitor 0.52-c: one cycle every 5 s | blue = decision (bits + state, make check) | green = matrix / amount | details: charge_flow.pdf, discharge_flow.pdf

inputs (bm_parse: only valid numbers, else stale)
PCC, Sofar Bat1, PV, heat pump W, R290 Hz, BMS SoC / U / CCL / DCL / lowest cell, season, forecast

charge path

D4 forecast export cap (May-Aug)
while the forecast expects export > 18 kW:
wait below the cap, charge only above it

S0 caps per stack: chg_cap, dis_cap (charger, BMS CCL / DCL)

D1 forecast, B1FIRST
-> S2 heat pump part (z2 = PCC + WP not covered)
-> S3 source DAY / B1 / STK / B1W

D5 PCC > 20 kW for 2 min (at once > 23 kW) -> DO4 pulse
pv_relay/DO4 (FoxESS shedding)
first: R290 boost + air conditioning at ≥ 18 kW
grid offline: AC-out feed > 5 kW / unit for 10 s -> at once

block

SoC

D2 mode per stack, first match:
BMS_MISSING > FORCE > STALE > LADESPERRE > CHARGE / FULL > PROT > CHARGING > DIS
FORCE = SoC < SOC_FORCE or LOW (DCL 0 / cell < 3.00 V, 0.52-c)
make check: 16128 combinations per stack (84 bit sets x 12 global x 16 states) -> mode:
BMS_MISSING 8064 · FORCE 5184 · STALE 1440 · LADESPERRE 360 · PROT 324 ·
CHARGE 288 · CHARGING 198 · DIS 198 · FULL 72

D-lead: stack more than SOC_BALANCE_ON ahead

src, z2

mode matrix BM_MODE_RULE
0 W | FORCE | charge | discharge

CHARGE

FORCE

DIS

charge amount
max(0, own + KP x (z2 + Bat1_eff - target))
target + EXPORT_CAP while D4 cap; ≤ sum of chg_cap

FORCE: min(500 W, chg_cap)
any time of day, no TOU window

alloc_levels up to chg_cap
stack by stack in CHARGE_PRIORITY,
lead stack last

0 W rows

discharge branch

BM_SRC_RULE[src]
STK: Bat1 signed, +trickle, idle 0
else: Bat1 charge only, idle 20 W

D3 PROP / IDLE

discharge amount
min(PROP ? deficit : idle, W_MAX, WP_CAP)

alloc_levels up to dis_cap
lead stack first, else
every stack the same share

setpoints per phase -> Multis (+ charge / - discharge)
rule text: mode WP: SRC: split [CCL / DCL / WPCAP]